



**RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF TECHNOLOGY**

**Bachelor of Information and Communication Technology
First Year – Semester I Examination – March 2024**

ICT 1305 – PROGRAM DESIGNING AND PROGRAMMING (Practical)

Index No.:

Time: Two (02) hours

- Answer **ALL** Questions
- Create a folder with your **index number** on the desktop and save your works in separate files named with the question numbers
- There are **EIGHT (08)** questions printed on **TWO (02)** pages.

1. Write a C program to check whether a number entered by a user is divisible by 7 or not.

Mark:	0	1	2	3	4	5	6	7	8	9	10
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2. Write a C program to convert Fahrenheit ($^{\circ}\text{F}$) into Celsius ($^{\circ}\text{C}$). Take the temperature in Fahrenheit as the user input.

$$\text{Celsius} = (5/9) * (\text{Fahrenheit} - 32)$$

Mark:	0	1	2	3	4	5	6	7	8	9	10
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3. Write a C program to get the distance traveled by a vehicle as a user input and calculate the total fare according to the given condition:

For the first 50 kilometers, Rs. 3 per kilometer

For the next 100 kilometers, Rs. 5 per kilometer

For any additional kilometers beyond 150, Rs. 7 per kilometer

Mark:	0	1	2	3	4	5	6	7	8	9	10
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4. Write a C program to display following pattern. Get the number of rows as user input.

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1
2 2
3 3 3
4 4 4 4

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Mark:	0	1	2	3	4	5	6	7	8	9	10
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5. Write a program in C to display the sum of integers from 1 to n. Take the n as a user input.

Mark:	0	1	2	3	4	5	6	7	8	9	10
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6. Write a C program that takes 10 integers from the user and adds all the positive numbers entered by the user. Neglect any negative numbers entered by the user.

Mark:	0	1	2	3	4	5	6	7	8	9	10
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7. Write a C program that defines a function to compute and display the multiplication of the integers from 1 to n, where n is taken as user input. Demonstrate the usage of this function in the main method.

Mark:	0	1	2	3	4	5	6	7	8	9	10
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8. Define a structure named "Triangle" with members; side1, side2 and side3. Write a C program to input data for a triangle, display its information, and find the perimeter of the triangle.

Mark:	0	1	2	3	4	5	6	7	8	9	10
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Total	/ 80
Final	/ 100